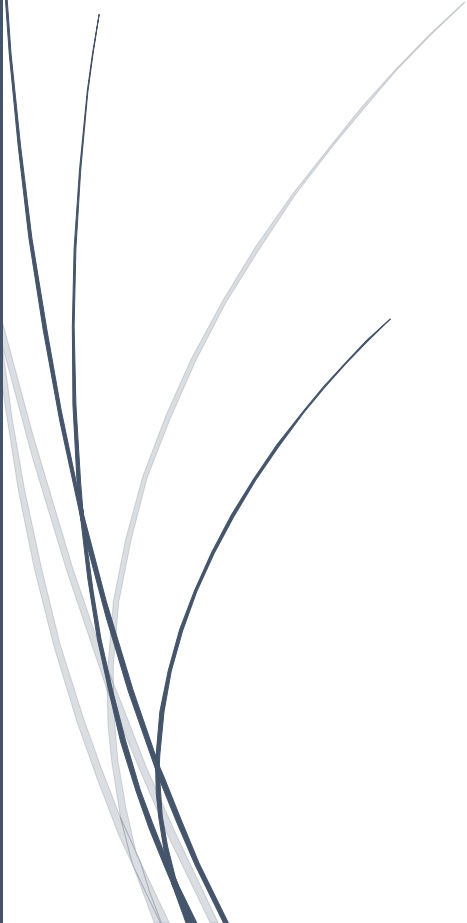


A dark blue vertical bar runs down the left side of the page. A blue arrow points to the right from this bar, containing the date.

11-1-2026

HIDS

Práctica 3-3

Several thin, curved lines in dark blue and light grey originate from the bottom left corner and sweep upwards and to the right.

SAVU, RALUCA ALEXANDRA
CIPFP MISLATA

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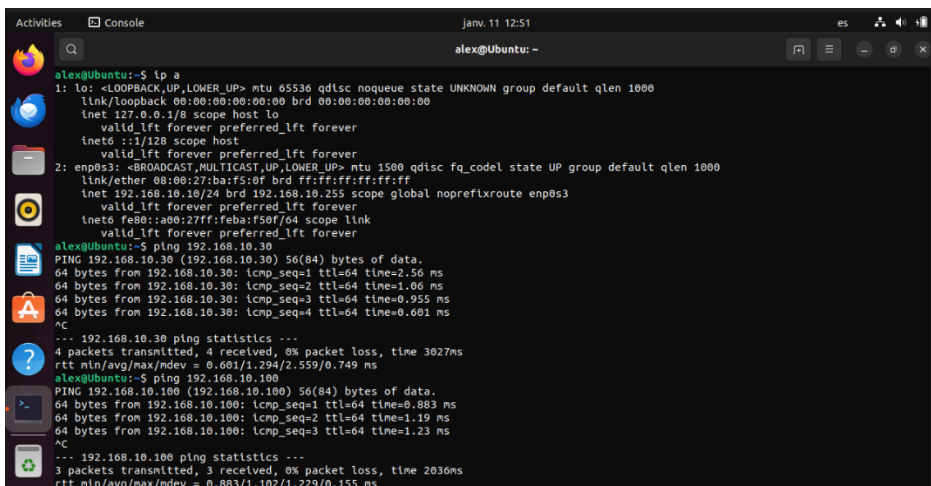
1. Máquinas

Para la realización de la práctica se ha optado por un escenario reducido de 3 máquinas virtuales, instalando el agente únicamente en una máquina Linux. El entorno está compuesto por un servidor Wazuh Manager (Ubuntu 22.04), una máquina Linux monitorizada (Ubuntu 22.04) y una máquina atacante con Kali Linux.

Las 3 máquinas tienen la Red en modo 'Red interna' para crear una red privada entre ellas.

1.1. Máquina Wazuh Manager

La máquina del Wazuh Manager es la versión Ubuntu 22.04 con una ip configurada a 192.168.10.10/24. En esta captura podemos ver la ip y la conexión con las otras máquinas.

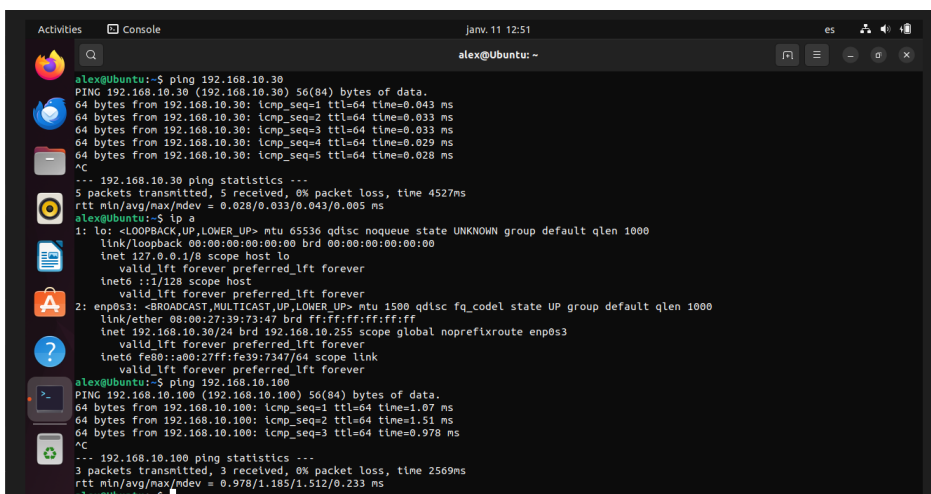


```

alex@Ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ba:f5:8f brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.10/24 brd 192.168.10.255 scope global noprefroute enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feba:f58f/64 scope link
        valid_lft forever preferred_lft forever
alex@Ubuntu:~$ ping 192.168.10.30
PING 192.168.10.30 (192.168.10.30) 56(84) bytes of data.
64 bytes from 192.168.10.30: icmp_seq=1 ttl=64 time=2.56 ms
64 bytes from 192.168.10.30: icmp_seq=2 ttl=64 time=1.06 ms
64 bytes from 192.168.10.30: icmp_seq=3 ttl=64 time=0.955 ms
64 bytes from 192.168.10.30: icmp_seq=4 ttl=64 time=0.601 ms
^C
--- 192.168.10.30 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3027ms
rtt min/avg/max/mdev = 0.601/1.294/2.559/0.749 ms
alex@Ubuntu:~$ ping 192.168.10.100
PING 192.168.10.100 (192.168.10.100) 56(84) bytes of data.
64 bytes from 192.168.10.100: icmp_seq=1 ttl=64 time=0.883 ms
64 bytes from 192.168.10.100: icmp_seq=2 ttl=64 time=1.19 ms
64 bytes from 192.168.10.100: icmp_seq=3 ttl=64 time=1.23 ms
^C
--- 192.168.10.100 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2036ms
rtt min/avg/max/mdev = 0.883/1.102/1.229/0.155 ms
  
```

1.2. Agente Linux

El agente Linux también es una máquina Ubuntu 22.04 con una ip esta vez configurada a 192.168.10.30/24. Comprobaremos la conexión con las otras máquinas:

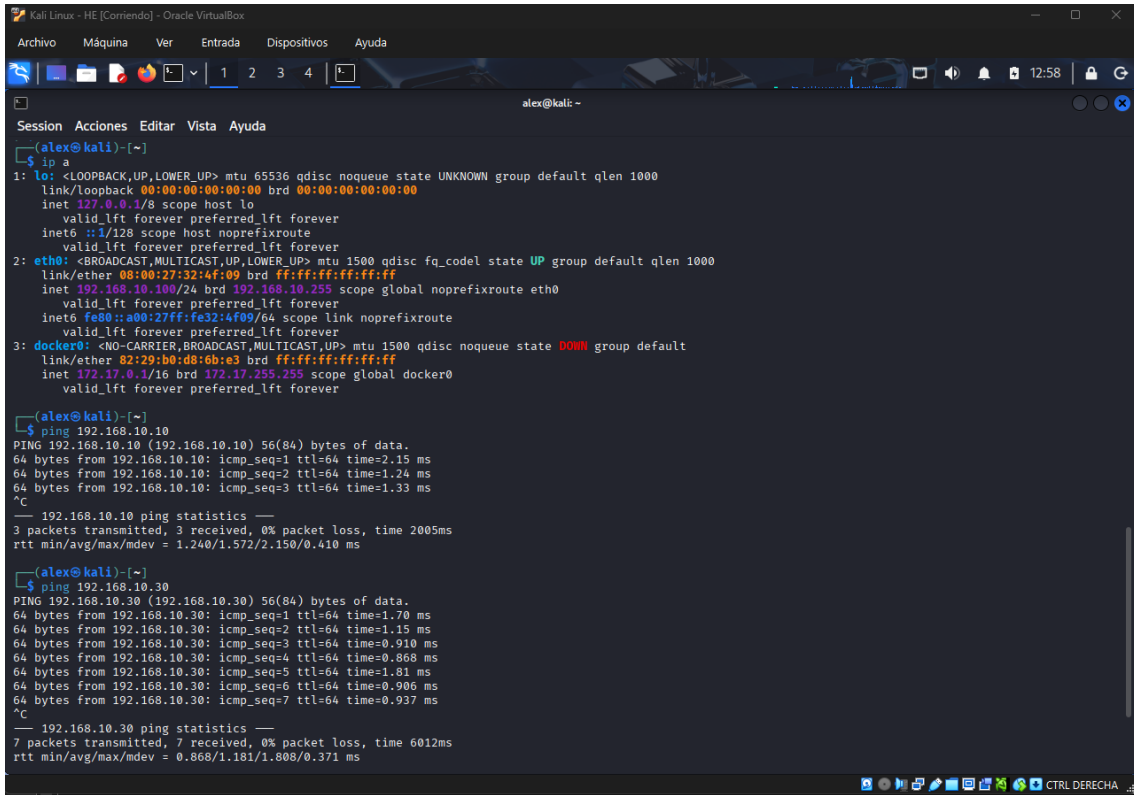


```

alex@Ubuntu:~$ ping 192.168.10.30
PING 192.168.10.30 (192.168.10.30) 56(84) bytes of data.
64 bytes from 192.168.10.30: icmp_seq=1 ttl=64 time=0.043 ms
64 bytes from 192.168.10.30: icmp_seq=2 ttl=64 time=0.033 ms
64 bytes from 192.168.10.30: icmp_seq=3 ttl=64 time=0.033 ms
64 bytes from 192.168.10.30: icmp_seq=4 ttl=64 time=0.029 ms
64 bytes from 192.168.10.30: icmp_seq=5 ttl=64 time=0.028 ms
^C
--- 192.168.10.30 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4527ms
rtt min/avg/max/mdev = 0.028/0.033/0.043/0.005 ms
alex@Ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:39:73:47 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.30/24 brd 192.168.10.255 scope global noprefroute enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe39:7347/64 scope link
        valid_lft forever preferred_lft forever
alex@Ubuntu:~$ ping 192.168.10.100
PING 192.168.10.100 (192.168.10.100) 56(84) bytes of data.
64 bytes from 192.168.10.100: icmp_seq=1 ttl=64 time=1.07 ms
64 bytes from 192.168.10.100: icmp_seq=2 ttl=64 time=1.51 ms
64 bytes from 192.168.10.100: icmp_seq=3 ttl=64 time=0.978 ms
^C
--- 192.168.10.100 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2569ms
rtt min/avg/max/mdev = 0.978/1.185/1.512/0.233 ms
alex@Ubuntu:~$
  
```

1.3. Atacante - Kali Linux

Esta última máquina es con la que atacaremos al agente Linux y tiene una IP de 192.168.10.100/24 y está en la misma red interna que las otras dos.



```

Kali Linux - HE [Corriendo] - Oracle VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda

alex@kali: ~
Session Acciones Editar Vista Ayuda

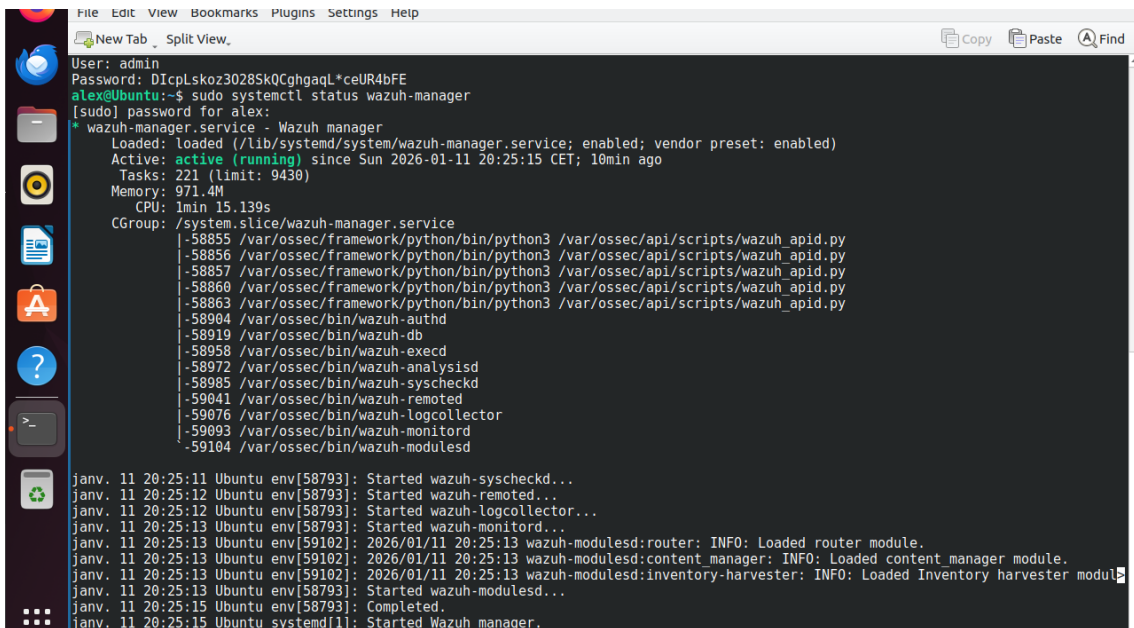
alex@kali:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:32:4f:09 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.100/24 brd 192.168.10.255 scope global noprefixroute eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe32:4f09/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 82:29:b0:d8:6b:e3 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever

alex@kali:~$ ping 192.168.10.10
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.
64 bytes from 192.168.10.10: icmp_seq=1 ttl=64 time=2.15 ms
64 bytes from 192.168.10.10: icmp_seq=2 ttl=64 time=1.24 ms
64 bytes from 192.168.10.10: icmp_seq=3 ttl=64 time=1.33 ms
^C
--- 192.168.10.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2005ms
rtt min/avg/max/mdev = 1.240/1.572/2.150/0.410 ms

alex@kali:~$ ping 192.168.10.30
PING 192.168.10.30 (192.168.10.30) 56(84) bytes of data.
64 bytes from 192.168.10.30: icmp_seq=1 ttl=64 time=1.70 ms
64 bytes from 192.168.10.30: icmp_seq=2 ttl=64 time=1.15 ms
64 bytes from 192.168.10.30: icmp_seq=3 ttl=64 time=0.910 ms
64 bytes from 192.168.10.30: icmp_seq=4 ttl=64 time=0.868 ms
64 bytes from 192.168.10.30: icmp_seq=5 ttl=64 time=1.81 ms
64 bytes from 192.168.10.30: icmp_seq=6 ttl=64 time=0.906 ms
64 bytes from 192.168.10.30: icmp_seq=7 ttl=64 time=0.937 ms
^C
--- 192.168.10.30 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6012ms
rtt min/avg/max/mdev = 0.868/1.181/1.808/0.371 ms
  
```

2. Instalación Wazuh-manager

Procedemos a instalar el wazuh manager en una de las máquinas Ubuntu según la guía.



```

File Edit View Bookmarks Plugins Settings Help
New Tab Split View Copy Paste Find

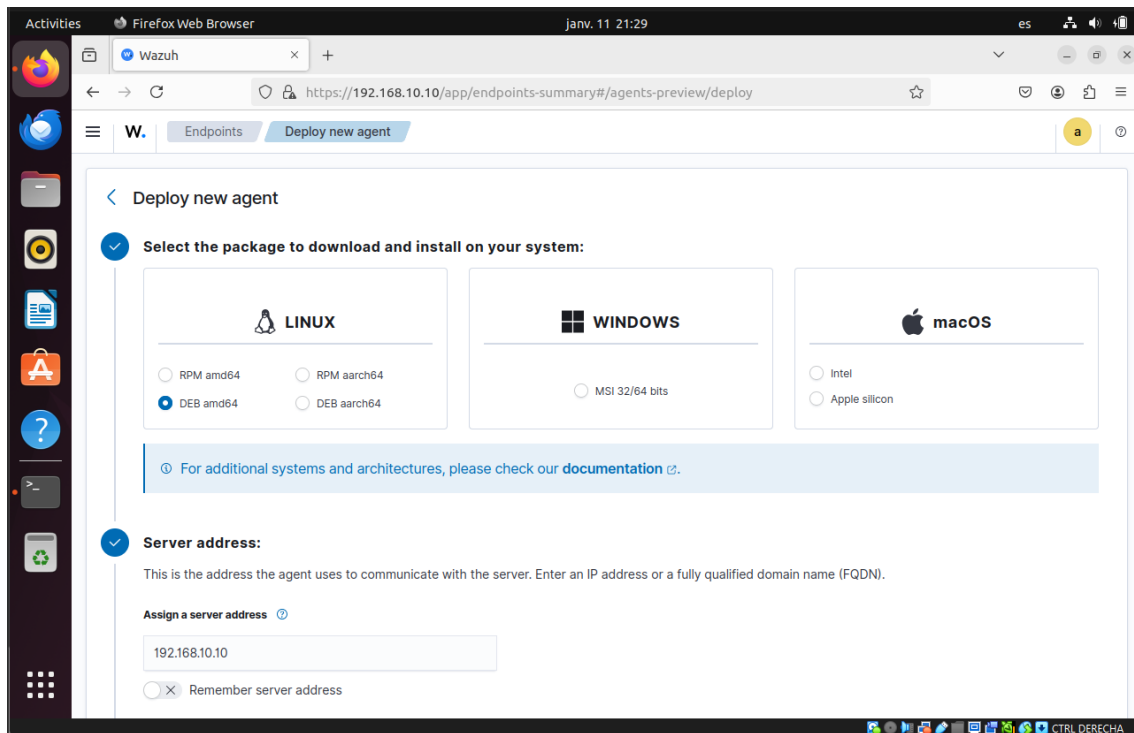
User: admin
Password: DICpLskoz3028SKQCghgaQL*ceUR4bFE
alex@Ubuntu:~$ sudo systemctl status wazuh-manager
[sudo] password for alex:
* wazuh-manager.service - Wazuh manager
   Loaded: loaded (/lib/systemd/system/wazuh-manager.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2026-01-11 20:25:15 CET; 10min ago
     Tasks: 221 (limit: 9430)
    Memory: 971.4M
           CPU: 1min 15.130s
    CGroup: /system.slice/wazuh-manager.service
            ┌─58855 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
            ┌─58856 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
            ┌─58857 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
            ┌─58860 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
            ┌─58863 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
            ┌─58904 /var/ossec/bin/wazuh-authd
            ┌─58910 /var/ossec/bin/wazuh-db
            ┌─58958 /var/ossec/bin/wazuh-execd
            ┌─58972 /var/ossec/bin/wazuh-analysisd
            ┌─58985 /var/ossec/bin/wazuh-syscheckd
            ┌─59041 /var/ossec/bin/wazuh-remoted
            ┌─59076 /var/ossec/bin/wazuh-logcollector
            ┌─59093 /var/ossec/bin/wazuh-monitord
            └─59104 /var/ossec/bin/wazuh-modulesd

janv. 11 20:25:11 Ubuntu env[58793]: Started wazuh-syscheckd...
janv. 11 20:25:12 Ubuntu env[58793]: Started wazuh-remoted...
janv. 11 20:25:12 Ubuntu env[58793]: Started wazuh-logcollector...
janv. 11 20:25:13 Ubuntu env[58793]: Started wazuh-monitord...
janv. 11 20:25:13 Ubuntu env[59102]: 2026/01/11 20:25:13 wazuh-modulesd:router: INFO: Loaded router module.
janv. 11 20:25:13 Ubuntu env[59102]: 2026/01/11 20:25:13 wazuh-modulesd:content manager: INFO: Loaded content manager module.
janv. 11 20:25:13 Ubuntu env[59102]: 2026/01/11 20:25:13 wazuh-modulesd:inventory-harvester: INFO: Loaded inventory harvester module.
janv. 11 20:25:13 Ubuntu env[58793]: Started wazuh-modulesd...
janv. 11 20:25:15 Ubuntu env[58793]: Completed.
janv. 11 20:25:15 Ubuntu systemd[1]: Started Wazuh manager.
  
```

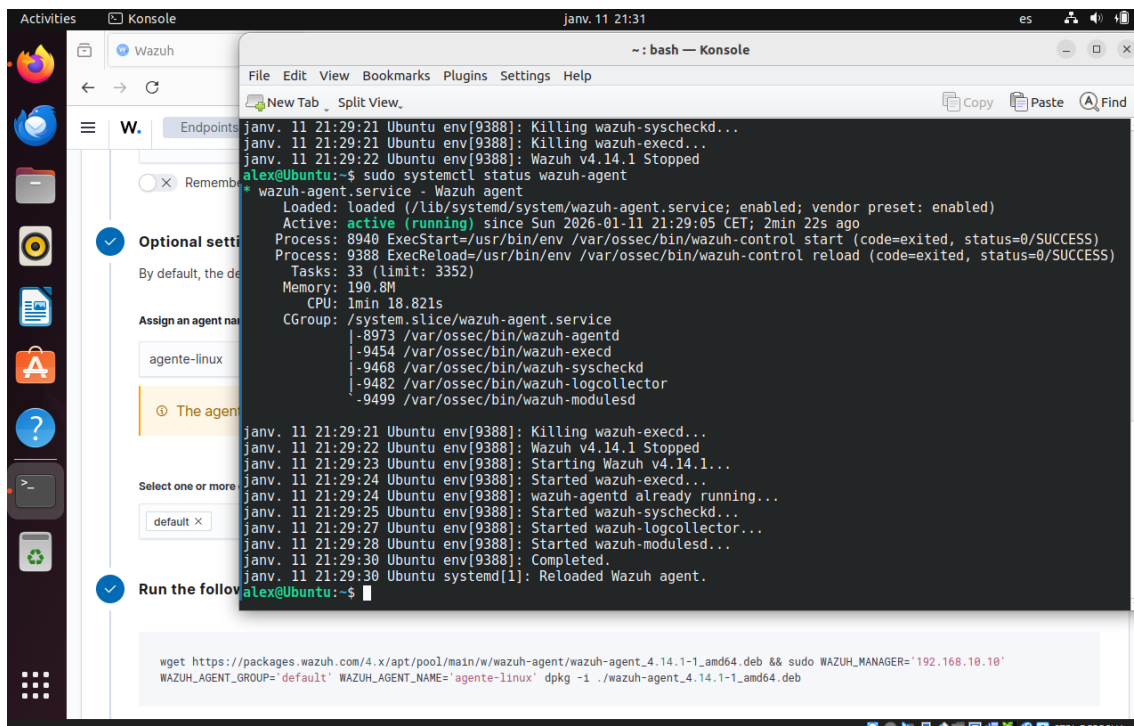
3.Instalación del agente linux

Ahora nos conectamos al mánager de wazuh desde la otra maquina linux para así instalar el agente wazuh. Utilizaremos la ip del servidor wazuh para conectarnos al dashboard y crear el agente.

<https://192.168.10.10:443>



Después de instalarlo en nuestra máquina agente podemos ver su estado:



También lo podemos ver en el dashboard:

Agents (1)								
status=active WQL								
☐	ID ↑	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status
☐	001	agente-linux	192.168.10.30	default	Ubuntu 22.04.5 LTS	node01	v4.14.1	active
Rows per page: 10 < 1 >								

4. Exploración

Ahora, desde la máquina kali Linux, probaremos a lanzar ataques hacia nuestro agente.

4.1. Ataque con Hydra

Con este comando, Hydra genera muchos intentos fallidos de login sshd. Así podemos comprobar como Wazuh detecta intentos fallidos, patrones de fuerza bruta y posibles ataques.

```
hydra -l root -P /usr/share/wordlists/metasploit/unix_passwords.txt ssh://192.168.10.30
```

```
(alex@kali)~$ hydra -l root -P /usr/share/wordlists/metasploit/unix_passwords.txt ssh://192.168.10.30
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-01-11 21:40:30
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 1021 login tries (l:1/p:1021), ~64 tries per task
[DATA] attacking ssh://192.168.10.30:22/
[STATUS] 235.00 tries/min, 235 tries in 00:01h, 786 to do in 00:04h, 16 active
[STATUS] 244.67 tries/min, 734 tries in 00:03h, 287 to do in 00:02h, 16 active
[STATUS] 251.25 tries/min, 1005 tries in 00:04h, 16 to do in 00:01h, 16 active
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-01-11 21:44:39
```

Podemos ver en el dashboard alertas de tipo “sshd:authentication failed” y ‘brute force’ gracias al ataque con Hydra:

Export Formatted Reset view 640 available fields Columns Density 1 fields sorted Full screen				
timestamp	agent.name	rule.description	rule.level	rule.id
Jan 11, 2026 @ 21:43:42.722	agente-linux	sshd: authentication failed.	5	5760
Jan 11, 2026 @ 21:43:42.718	agente-linux	sshd: authentication failed.	5	5760
Jan 11, 2026 @ 21:43:42.714	agente-linux	sshd: authentication failed.	5	5760
Jan 11, 2026 @ 21:43:42.714	agente-linux	sshd: brute force trying to get access to the system. Authentication failed.	10	5763
Jan 11, 2026 @ 21:43:42.709	agente-linux	PAM: User login failed.	5	5503
Jan 11, 2026 @ 21:43:42.706	agente-linux	Multiple authentication failures.	10	40111
Jan 11, 2026 @ 21:43:42.686	agente-linux	syslog: User authentication failure.	5	2501
Jan 11, 2026 @ 21:43:42.686	agente-linux	Maximum authentication attempts exceeded.	8	5758
Jan 11, 2026 @ 21:43:42.686	agente-linux	syslog: User missed the password more than one time	10	2502
Jan 11, 2026 @ 21:43:42.645	agente-linux	syslog: User missed the password more than one time	10	2502
Jan 11, 2026 @ 21:43:42.641	agente-linux	Maximum authentication attempts exceeded.	8	5758
Jan 11, 2026 @ 21:43:42.641	agente-linux	syslog: User authentication failure.	5	2501
Jan 11, 2026 @ 21:43:40.757	agente-linux	PAM: User login failed.	5	5503
Jan 11, 2026 @ 21:43:40.700	agente-linux	syslog: User missed the password more than one time	10	2502
Jan 11, 2026 @ 21:43:40.692	agente-linux	syslog: User authentication failure.	5	2501

4.2. Vigilar etc/passwd

Ahora cambiaremos el archivo `/etc/passwd` y observamos la alerta que aparece en el Wazuh.

```
gdm:x:128:134:Gnome Display Manager:/var/lib/gdm3:/bin/false
alex:x:1000:1000:alex,,,:/home/alex:/bin/bash
sshd:x:129:65534::/run/sshd:/usr/sbin/nologin
wazuh-indexer:x:999:999:wazuh-indexer user/group:/home/wazuh-indexer
wazuh-dashboard:x:131:138::/nonexistent:/bin/false
wazuh:x:130:137::/var/ossec:/sbin/nologin
# Comentario

```

Aquí podemos ver en el dashboard como salta la alerta de la modificación del archivo passwd:

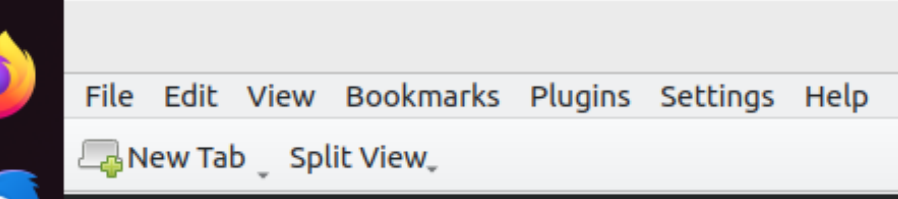
```

    rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV.32.2 rule.firedtimes: 6 rule.mitre.technique: Valid Accounts
  > Jan 11, 2026 @ 22:06:46.414 predecoder.hostname: Ubuntu predecoder.program_name: sudo predecoder.timestamp: Jan 11 21:06:44 input.type: log
    agent.ip: 192.168.10.30 agent.name: agente-linux agent.id: 001 manager.name: Ubuntu data.srcuser: alex
    data.dstuser: root data.tty: pts/0 data.pwd: /home/alex data.command: /usr/bin/nano /etc/passwd rule.mail: false
    rule.level: 3 rule.pci_dss: 10.2.5, 10.2.2 rule.hipaa: 164.312.b rule.tsc: CC6.8, CC7.2, CC7.3 rule.description:
    Successful sudo to ROOT executed. rule.groups: syslog, sudo rule.nist_800_53: AU.14, AC.7, ac.6 rule.gdpr: IV.32.2
  > Jan 11, 2026 @ 22:06:48.398 predecoder.hostname: Ubuntu predecoder.program_name: sudo predecoder.timestamp: Jan 11 21:06:38 input.type: log

```

4.3. Alerta por cambio de permisos

Ahora en la maquina agente cambiaremos los permisos de unos archivos y vemos las alertas que saltan en el Wazuh, ya que tenemos implementadas alertas si se cambia cualquier archivo del directorio `/etc/`.



The screenshot shows a Linux desktop environment. On the left is a vertical dock with icons for Firefox, Telegram, a file manager, and a music player. The top of the screen features a dark header bar with the text "Activities" and a window icon labeled "Konsole". The main area is occupied by a terminal window titled "Konsole". The terminal has a menu bar with "File", "Edit", "View", "Bookmarks", "Plugins", "Settings", and "Help". Below the menu bar is a toolbar with a "New Tab" button and a "Split View" button. The terminal's command history is as follows:

```
alex@Ubuntu:~$ sudo nano /etc/passwd
[sudo] password for alex:
alex@Ubuntu:~$ sudo nano /etc/passwd
alex@Ubuntu:~$ sudo chmod 777 /bin/ls
alex@Ubuntu:~$ sudo chmod 755 /bin/ls
alex@Ubuntu:~$
```

The prompt "alex@Ubuntu:~\$" is shown in green text. The terminal background is dark gray, and the text is white except for the prompt which is green.

U.14, AC.7 rule.gpg13: 7.8, 7.9 rule.gdpr: IV_32.2 location: journald decoder.parent: pam decoder.name: pam id:
<pre> > Jan 11, 2026 @ 22:12:23.153 predecoder.hostname: Ubuntu predecoder.program_name: sudo predecoder.timestamp: Jan 11 21:12:21 input.type: log agent.ip: 192.168.10.30 agent.name: agente-linux agent.id: 001 manager.name: Ubuntu data.srcuser: alex data.dstuser: root data.tty: pts/0 data.pwd: /home/alex data.command: /usr/bin/chmod 755 /bin/lis rule.mail: false rule.level: 3 rule.pci_dss: 10.2.5, 10.2.2 rule.hipaa: 164.312.b rule.tsc: CC6.8, CC7.2, CC7.3 rule.description: Successful sudo to ROOT executed. rule.groups: syslog, sudo rule.nist_800_53: AU.14, AC.7, AC.6 rule.gdpr: IV_32.2 </pre>
<pre> > Jan 11, 2026 @ 22:12:23.153 predecoder.hostname: Ubuntu predecoder.program_name: sudo predecoder.timestamp: Jan 11 21:12:21 input.type: log </pre>

4.4. Alerta por intentos fallidos de login manual (SSH)

Ahora, desde la maquina Kali Linux intentamos conectarnos por ssh a la máquina agente con una contraseña incorrecta para así comprobar el tipo de alertas en el Wazuh.

```

(alex@kali)-[~]
$ ssh alex@192.168.10.30
The authenticity of host '192.168.10.30 (192.168.10.30)' can't be established.
ED25519 key fingerprint is: SHA256:jBh8CEkvJ9sJ7IPD4XWVTQ99zwfxD2oy80ywwq15vFQ
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.10.30' (ED25519) to the list of known hosts.
alex@192.168.10.30's password:
Permission denied, please try again.
alex@192.168.10.30's password:
Permission denied, please try again.
alex@192.168.10.30's password:
alex@192.168.10.30: Permission denied (publickey,password).
(alex@kali)-[~]

```

Ahora podemos ver que la alerta es de tipo 'sshd: authentication_failed':

timestamp	agent.name	rule.description	rule.level	rule.id
Jan 11, 2026 @ 22:25:37.305	agente-linux	syslog: User missed the password more than one time	10	2502
Jan 11, 2026 @ 22:25:37.303	agente-linux	sshd: authentication failed.	5	5760
Jan 11, 2026 @ 22:25:31.297	agente-linux	sshd: authentication failed.	5	5760
Jan 11, 2026 @ 22:25:27.294	agente-linux	sshd: authentication failed.	5	5760

****** A diferencia del ataque por fuerza bruta, no vemos ni la alerta de 'Maximun Authentication attempts exceeded' ni la de 'brute force', ya que son intentos manuales de login.

5. Conclusión

Durante la práctica se comprobó cómo Wazuh es capaz de detectar distintos tipos de ataques en función de su comportamiento. Los intentos fallidos realizados de forma manual generan alertas individuales de bajo nivel, mientras que los ataques automatizados son identificados mediante reglas, produciendo alertas de mayor severidad. Asimismo, la monitorización de la integridad de archivos y los cambios en permisos binarios críticos permiten detectar modificaciones no autorizadas en el sistema. Esto demuestra la eficacia de Wazuh como sistema HIDS para la detección y análisis de eventos de seguridad en tiempo real.

